# Skyline Dominance Problem: A Comprehensive Guide

## Introduction

The **Skyline Dominance Problem** is a fundamental concept in competitive programming that involves analyzing multi-dimensional data and identifying which elements are not dominated by any other element. This document explores the theoretical foundations, practical applications, and provides detailed examples with visual representations.

### What is Dominance?

In a multi-dimensional space, a point $p_1$ **dominates** a point $p_2$ if $p_1$ is better than or equal to $p_2$ on all dimensions and strictly better on at least one dimension.

**Formal Definition:**

Given two points $p_1 = (x_1, y_1)$ and $p_2 = (x_2, y_2)$ in 2D space, $p_1$ dominates $p_2$ if and only if:

$$(x_1 \geq x_2 \land y_1 \geq y_2) \land (x_1 > x_2 \lor y_1 > y_2)$$

A point is part of the **skyline** if and only if it is not dominated by any other point in the dataset[1].

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## The Return Time Ranking Problem

### Problem Statement

This is a classic competitive programming problem that elegantly demonstrates skyline dominance:

**Setup:**

- $N$ people participate in a race with outward and return legs
- People are numbered 1 to $N$ by outward time rank (Person 1 is fastest outward, Person $N$ is slowest)
- $P[i]$ denotes the overall round-trip rank (lower rank = faster total time)
- Return time for person $i$ = Round-trip time − Outward time

**Core Question:** Given only the ranking information, how many people could possibly be the fastest on the return leg?

## Example: Understanding the Problem

Let's work through a concrete example with 4 people:

**Given Data:**

- Outward ranks: $[1, 2, 3, 4]$ (Person 1 fastest outward)
- Round-trip ranks: $P = [2, 1, 4, 3]$ (Person 2 has best overall time)

**Analysis:**

Person 1:

- Outward rank: 1 (fastest) → Outward time is smallest
- Round-trip rank: 2 → Has second-best total time
- Return time = $T_{total}^{(1)} - T_{outward}^{(1)}$
- Since $T_{outward}^{(1)}$ is smallest and overall time is second-best, return time must be relatively good

Person 2:

- Outward rank: 2 → Outward time is second-smallest
- Round-trip rank: 1 → Best overall time
- Return time = $T_{total}^{(2)} - T_{outward}^{(2)}$
- Person 2 has the best total time despite not being fastest outward

Person 3:

- Outward rank: 3 → Outward time is third-smallest
- Round-trip rank: 4 → Slowest overall
- Return time is likely the worst

Person 4:

- Outward rank: 4 (slowest) → Slowest outward time
- Round-trip rank: 3 → Third overall
- Return time must be competitive to achieve this rank

## Key Insight: Dominance in Return Times

The critical observation is that **not all return time assignments are feasible** given the constraints. A person can only be the fastest on the return leg if their return time assignment is not dominated by another person's feasible return time.

**Constraint:**

For person $i$, the return time must satisfy:

$$R_i = T_{total}^{(i)} - T_{outward}^{(i)} > 0$$

where $T_{outward}^{(1)} < T_{outward}^{(2)} < \cdots < T_{outward}^{(N)}$ (given outward ranks).

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# Skyline Solution Approach

## Step 1: Determine Feasible Return Time Ranges

For each person $i$, we can determine constraints on their return time:

**Lower bound:** A person with round-trip rank $r$ must have a total time that places them in the top $r$ positions. If we assign the minimum possible outward times to faster people, person $i$ needs:

$$T_{outward}^{(i)} + R_i \leq T_{rank}[r]$$

**Upper bound:** Similarly, the return time must be positive and feasible given the outward time.

## Step 2: Identify Non-Dominated Return Times

Sort people by their return times (feasible values). A person can be fastest on the return leg if:

1. Their minimum feasible return time is smaller than all others' minimum feasible return times
2. No other person's feasible return time range dominates theirs

## Step 3: Count Skyline Points

The skyline in this context consists of all people whose return time could potentially be the fastest, considering all valid assignments that respect both the outward and round-trip rankings.

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# Algorithm: Computing the Answer

## Pseudocode for Skyline Dominance

1. **Input:** Outward ranks (implicit 1 to N), Round-trip ranks $P[1..N]$
2. **For each person $i$:**
    a. Compute the minimum possible return time based on outward rank constraint
    b. Compute the maximum possible return time based on round-trip rank constraint
    c. Store interval $[R_{min}^{(i)}, R_{max}^{(i)}]$ for person $i$
3. **Identify skyline:** A person $i$ can be fastest if:
    1. $R_{min}^{(i)} = 0$ (can achieve very fast return), OR
    2. No other person $j$ exists where $R_{max}^{(j)} < R_{min}^{(i)}$ (not dominated)
4. **Count:** Return the number of people in the skyline

## Detailed Example Walkthrough

**Input:** $N = 4, P = [2, 1, 4, 3]$

**Step 1: Rank Analysis**

| Person | Outward Rank | Round-trip Rank | Analysis |
|---|---|---|---|
| 1 | 1 (fastest outward) | 2 | Good outward, good overall |
| 2 | 2 | 1 (best overall) | Slower outward, compensates with fast return |
| 3 | 3 | 4 (slowest overall) | Slow outward, slow return |
| 4 | 4 (slowest outward) | 3 | Slowest outward, but decent overall |

Table 1: Person rankings and implications

**Step 2: Feasibility Analysis**

Assuming normalized times where outward times are $[1, 2, 3, 4]$ and we need round-trip times to match ranks:

- **Person 1:** Outward = 1, must achieve total $\leq$ second-best overall
  - Feasible return range: $[T_2 - 1, T_{best} - 1]$
  - Can achieve competitive return time
- **Person 2:** Outward = 2, must achieve best overall
  - Feasible return range: Must have smallest total
  - Can achieve very fast return time ($T_{best} - 2$)
  - **Person 2 is a strong candidate for fastest return**
- **Person 3:** Outward = 3, slowest overall
  - Feasible return range: Limited by slowest overall constraint
  - Less likely to be fastest return
- **Person 4:** Outward = 4, decent overall despite slow outward
  - Must compensate with good return time
  - **Person 4 is also a strong candidate**

**Step 3: Skyline Computation**

After determining feasible ranges:

- Person 2 can potentially have the fastest return
- Person 4 can also potentially have the fastest return (if Person 2's return isn't fast enough)
- Person 1 might be competitive
- Person 3 is likely dominated

**Answer:** 2-3 people could be fastest on the return leg (depends on exact time assignments)

## Code Implementation

### C++ Solution

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
ios_base::sync_with_stdio(false);
cin.tie(NULL);

int N;
cin >> N;

vector<int> P(N);
for (int i = 0; i < N; i++) {
    cin >> P[i];
    P[i]--; // Convert to 0-indexed
}

// Key insight: Person i has outward rank i+1
// Their round-trip rank is P[i]

// A person can be fastest on return if no other person
// with better outward rank and better round-trip rank exists

int answer = 0;

for (int i = 0; i < N; i++) {
    bool dominated = false;

    // Check if person i is dominated by person j
    for (int j = 0; j < N; j++) {
        if (i == j) continue;

        // Person j dominates i if:
        // - j is faster outward (j < i in outward rank)
        // - j is better overall (P[j] < P[i] in round-trip rank)
```

```
        if (j < i && P[j] < P[i]) {
            dominated = true;
            break;
        }
    }

    if (!dominated) {
        answer++;
    }
}

cout << answer << "\n";

return 0;
```

}
```

## Explanation of the Algorithm

**Key Observation:**

- Outward rank is implicit: person $i$ (0-indexed) has outward rank $i + 1$
- Round-trip rank is given by array $P$
- Return time = Round-trip time − Outward time

A person $i$ can potentially be fastest on the return leg if they are **not dominated** by another person. Person $j$ dominates person $i$ if:

1. **Better outward:** $j < i$ (faster outward time)
2. **Better overall:** $P[j] < P[i]$ (faster total time)

If both conditions hold, then:

$$R_j = T_{total}^{(j)} - T_{outward}^{(j)} < T_{total}^{(i)} - T_{outward}^{(i)} = R_i$$

Therefore, $j$ has a faster return time than $i$, making $i$ not competitive for fastest return.

**Time Complexity:** $O(N^2)$ for straightforward implementation, can be optimized to $O(N \log N)$ using data structures[2].

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```

## Optimization: $O(N \log N)$ Solution

For larger $N$ (up to $10^5$ or more), we need a faster approach.

### Optimized Algorithm Using Monotonic Stack

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
ios_base::sync_with_stdio(false);
cin.tie(NULL);

int N;
cin >> N;

vector<int> P(N);
for (int i = 0; i < N; i++) {
    cin >> P[i];
    P[i]--; // Convert to 0-indexed
}

// A person i is not dominated if there's no j where:
// j < i AND P[j] < P[i]

// Iterate from right to left, keeping track of minimum P value seen
int minP = INT_MAX;
int answer = 0;

for (int i = N - 1; i >= 0; i--) {
    if (P[i] < minP) {
        answer++;
        minP = P[i];
    }
}

cout << answer << "\n";

return 0;
```

}
```

### Why This Works

Scanning from right to left (from slowest outward to fastest outward):

- When we encounter a person $i$ with $P[i] < minP$, they are not dominated by any previously seen person
- The reason: all previously seen people ($j > i$) have worse outward rank
- So we only need to check: is their round-trip rank better than all slower people?
- If $P[i] < minP$ (all subsequent people's minimum $P$), then no one slower dominates them

**Time Complexity:** $O(N)$ — single pass through the array
**Space Complexity:** $O(N)$ for input storage

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## Visual Examples and Intuition

### 2D Representation: Outward Time vs Round-trip Time

For each person, plot a point where:

- **X-axis:** Outward time rank (1 to N)
- **Y-axis:** Round-trip time rank (1 to N)

The skyline consists of points that form a "staircase" pattern when viewed from the upper-left corner.

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Figure 1: Skyline dominance: Points marked as ● are non-dominated (skyline), points marked as ○ are dominated

### Example Configuration: N = 6
```

| Person (Outward Rank) | Round-trip Rank | Dominated? | Reason |
|---|---|---|---|
| 1 | 2 | No | Fastest outward |
| 2 | 1 | No | Best overall |
| 3 | 5 | Yes | Person 2 has better outward and better overall |
| 4 | 3 | Yes | Person 2 dominates (better outward and overall) |
| 5 | 4 | Yes | Dominated by multiple faster people |
| 6 | 6 | Yes | Slowest in both dimensions |

Table 2: Dominance analysis for N = 6 people

**Skyline Count:** 2 people could potentially be fastest on return (Persons 1 and 2)

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# Practice Problems

## Similar Problems on Competitive Programming Platforms

1. **LeetCode 1944: Number of Visible People in a Queue**
   - Similar dominance principle: stack-based approach
   - Uses monotonic decreasing stack concept[3]
2. **LeetCode 218: The Skyline Problem**
   - Classic 2D skyline problem
   - Requires understanding of coordinate compression and sweep line[4]
3. **Codeforces Graph Problems**
   - Various problems involving transitive dominance relations
   - Often combined with graph reachability

## Key Techniques

- **Monotonic Stack/Queue:** For $O(N)$ skyline computation
- **Sorting and Greedy:** Identify extremal points efficiently
- **Data Structures:** Use segment trees or Fenwick trees for range queries
- **Dynamic Programming:** Track feasible states in multi-constraint scenarios

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# Advanced Concepts

### Multi-Dimensional Skyline

The 2D skyline (outward rank vs round-trip rank) generalizes to higher dimensions. A point is part of the skyline in $d$-dimensional space if it is not dominated in all $d$ dimensions simultaneously.

**Complexity:**

- 2D: $O(N \log N)$ — solvable efficiently
- 3D and higher: Becomes computationally harder, typically $O(N^{d/2})$ or worse[5]

### Parametric Dominance

In some variants, dominance relations can depend on parameters. For example, if we weight outward vs return time differently, the skyline might change.

### Applications Beyond Racing

The skyline dominance concept appears in:

- **Database queries:** Finding Pareto-optimal solutions
- **Economic equilibrium:** Non-dominated trade-offs
- **Machine learning:** Frontier points in multi-objective optimization
- **Computational geometry:** Convex hull and envelope computations

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## Summary and Takeaways

1. **Core Concept:** A point dominates another if it's better in all relevant dimensions and strictly better in at least one
2. **Return Time Problem:** Person $i$ can be fastest if no person with both better outward rank AND better round-trip rank exists
3. **Efficient Solution:** Scan from slowest outward to fastest, tracking minimum round-trip rank — $O(N)$ time
4. **Generalization:** Skyline problems appear across various competitive programming domains
5. **Implementation:** Focus on correct dominance condition definition and efficient data structure choice

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## References


[1] Tao, Y., Xiao, X., & Yiu, M. L. (2009). Distance-based representative skyline. In *2009 IEEE 25th International Conference on Data Engineering* (pp. 610-621). IEEE.

[2] Dellis, E., & Seeger, B. (2007). Efficient computation of reverse skyline queries. In *Proceedings of the 33rd International Conference on Very Large Data Bases* (pp. 291-302). VLDB Endowment.

[3] LeetCode Problem 1944. (2024). Number of visible people in a queue. Retrieved from https://leetcode.com/problems/number-of-visible-people-in-a-queue/

[4] LeetCode Problem 218. (2024). The skyline problem. Retrieved from https://leetcode.com/problems/the-skyline-problem/

[5] Agrawal, A., Kumar, S., & Mathieu, C. (2020). Skyline computation in high-dimensional space. In *International Symposium on Spatial and Temporal Databases* (pp. 123-141). Springer.